gdTAI v3 in GSE305372 CD8 T cells

Independent application to author-annotated lung and lung-associated lymph-node CD8 cells

157,846
CD8A cells analyzed

651
gdTAI-positive cells

0.412%
predicted fraction

0.936
registered threshold

579
predictions with TRDV expression

0.191%
paired TRA/TRB conflict-screening rate

Interpretation limit. These downloaded CD8 processed objects contain TRA/TRB metadata but no TRG/TRD clonotype fields. Therefore, this is an inference report, not a precision/recall evaluation. Model-positive cells with paired TRA/TRB evidence (245 of 128,446; 0.191%) are apparent alpha-beta-associated calls requiring review; they cannot be labeled definitive false positives without matched gamma-delta TCR evidence.

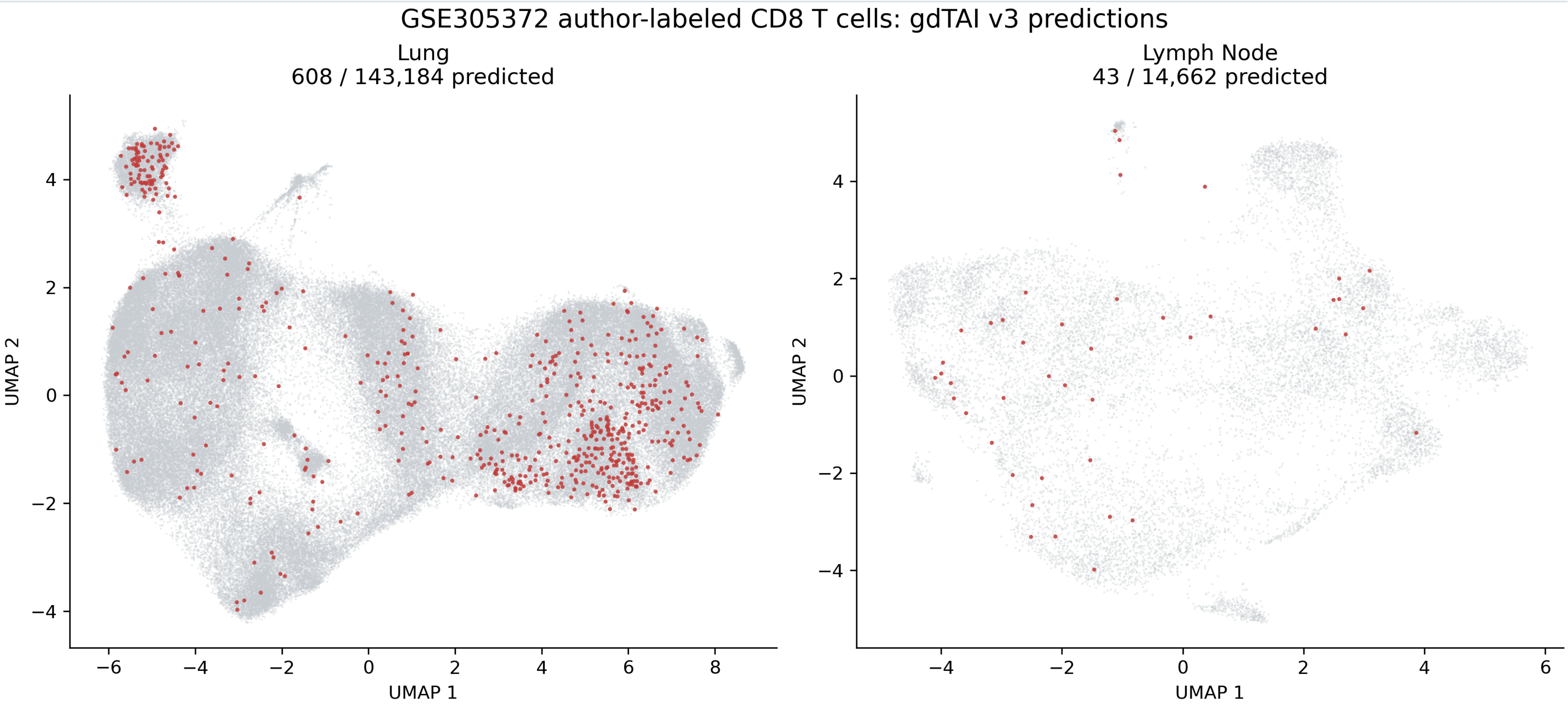
Input and method

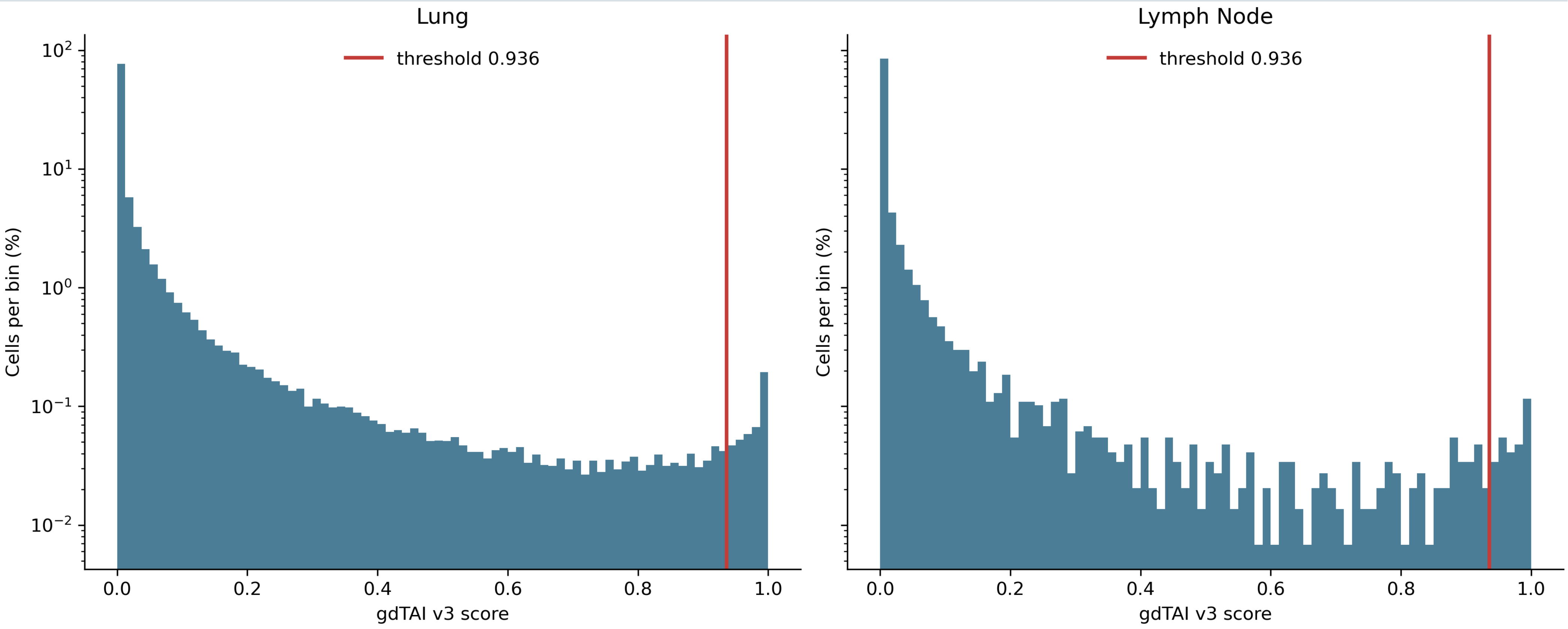
The analysis downloaded only the two GEO CD8 processed objects and retained cells with the authors' `cite.cell.type.tag == CD8A`. It did not select cells using TRD, TRG, gdTAI score, or any model feature. Raw RNA counts for the packaged 210 genes were normalized as `log1p(count * 10,000 / whole-transcriptome total counts)`. The promoted `gdtai_v3` Round 14 artifact was applied at its fixed threshold of `0.936`. This CD8A restriction means CD8-negative gamma-delta T cells are outside the analysis scope.

Model SHA256: `16dedc0081da9b8487887341232bcf8c9c9403dd3bbd72e04cab43d4cd7b2e09`

Overall results

scope	source_compartment	tissue	n_cells	n_predicted	predicted_fraction	median_score	p95_score
all_CD8A			157846	651	0.0041	0.0004	0.1566
	lung	lung	143184	608	0.0042	0.0004	0.1645
	lymph_node	lung-associated lymph node	14662	43	0.0029	0.0001	0.0816



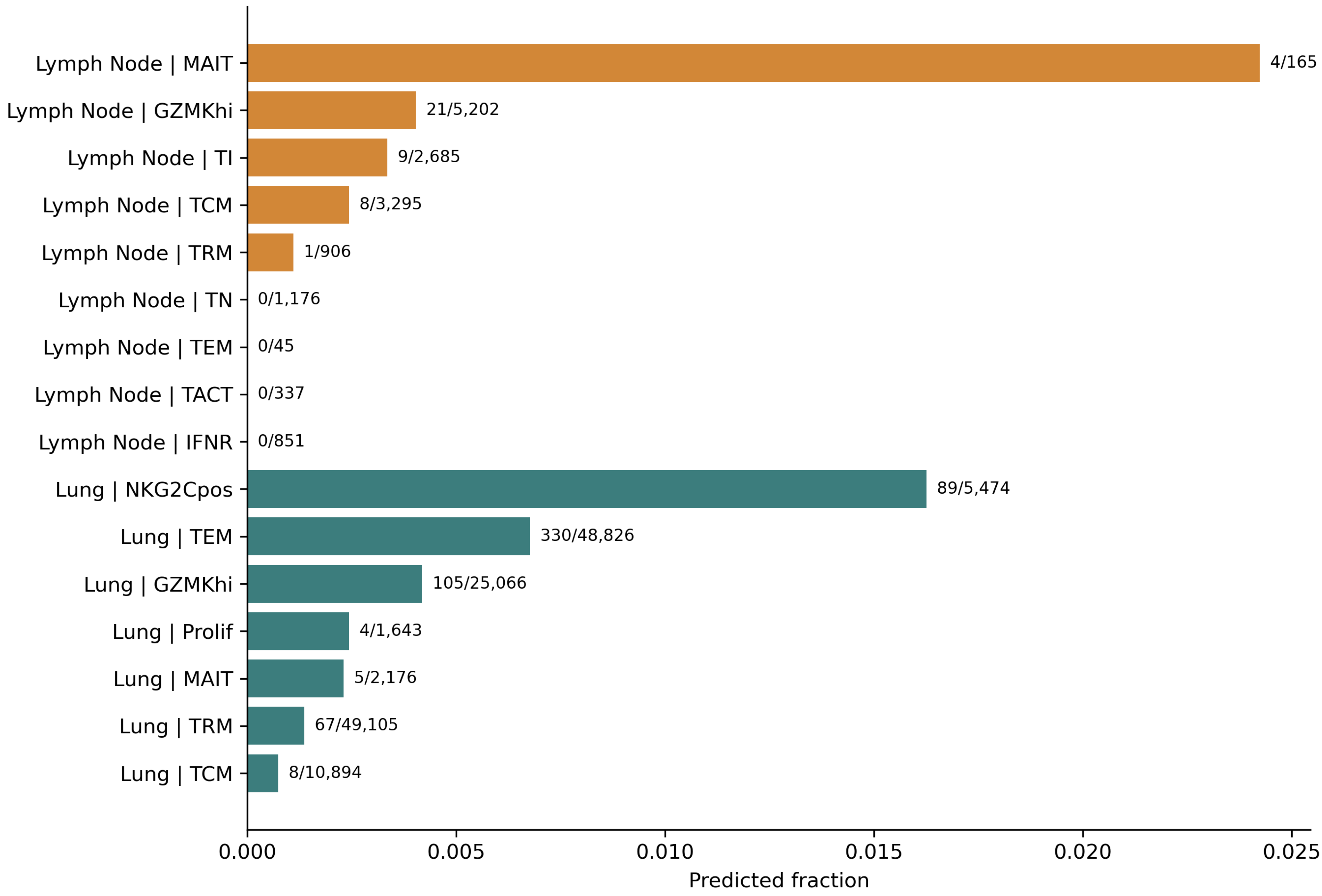


Author phenotype clusters

98 predictions fall in the authors' NKG2C-positive or MAIT clusters. These populations receive explicit attention because cytotoxic T cells, MAIT cells, gamma-delta T cells, and NK-like states can share expression programs.

source_compartment	author_cluster_id	author_cluster	n_cells	n_predicted	predicted_fraction	predicted_fraction_ci_low	predicted_fraction_ci_high
lung	0	TRM	49105	67	0.0014	0.0011	0.0017
lung	1	TEM	48826	330	0.0068	0.0061	0.0075
lung	2	GZMKhi	25066	105	0.0042	0.0035	0.0051
lung	3	TCM	10894	8	0.0007	0.0004	0.0014
lung	4	NKG2Cpos	5474	89	0.0163	0.0132	0.0200
lung	5	MAIT	2176	5	0.0023	0.0010	0.0054

source_compartment	author_cluster_id	author_cluster	n_cells	n_predicted	predicted_fraction	predicted_fraction_ci_low	predicted_fraction_ci_high
lung	6	Prolif	1643	4	0.0024	0.0009	0.0062
lymph_node	0	GZMKhi	5202	21	0.0040	0.0026	0.0062
lymph_node	1	TI	2685	9	0.0034	0.0018	0.0064
lymph_node	2	TCM	3295	8	0.0024	0.0012	0.0048
lymph_node	3	TN	1176	0	0.0000	0.0000	0.0033
lymph_node	4	IFNR	851	0	0.0000	0.0000	0.0045
lymph_node	5	TRM	906	1	0.0011	0.0002	0.0062
lymph_node	6	TACT	337	0	0.0000	0.0000	0.0113
lymph_node	7	MAIT	165	4	0.0242	0.0095	0.0607
lymph_node	8	TEM	45	0	0.0000	0.0000	0.0787


Lymph Node | MAIT
Lymph Node | GZMKhi
Lymph Node | TI
Lymph Node | TCM
Lymph Node | TRM
Lymph Node | TN
Lymph Node | TEM
Lymph Node | TACT
Lymph Node | IFNR
Lung | NKG2Cpos
Lung | TEM
Lung | GZMKhi
Lung | Prolif
Lung | MAIT
Lung | TRM
Lung | TCM
4/165
21/5,202
9/2,685
8/3,295
1/906
0/1,176
0/45
0/337
0/851
89/5,474
330/48,826
105/25,066
4/1,643
5/2,176
67/49,105
8/10,894
0.000
0.005
0.010
0.015
0.020
0.025
Predicted fraction

Donor audit

Calls are shown for every available author donor identifier so that donor concentration and small-denominator effects remain visible.

source_compartment	donor.id.tag	n_cells	n_predicted	predicted_fraction	predicted_fraction_ci_low	predicted_fraction_ci_high
lung	L003	11569	16	0.0014	0.0009	0.0022
lung	L004	5462	5	0.0009	0.0004	0.0021
lung	L005	7527	17	0.0023	0.0014	0.0036
lung	L006	1199	2	0.0017	0.0005	0.0061
lung	L007	6425	47	0.0073	0.0055	0.0097
lung	L008	2098	12	0.0057	0.0033	0.0100
lung	L009	6148	79	0.0128	0.0103	0.0160
lung	L010	9302	51	0.0055	0.0042	0.0072
lung	L011	91	0	0.0000	0.0000	0.0405
lung	L012	2637	28	0.0106	0.0074	0.0153
lung	L013	1081	0	0.0000	0.0000	0.0035
lung	L014	3827	8	0.0021	0.0011	0.0041
lung	L015	5103	30	0.0059	0.0041	0.0084
lung	L016	3895	29	0.0074	0.0052	0.0107
lung	L017	4044	2	0.0005	0.0001	0.0018
lung	L019	2	0	0.0000	0.0000	0.6576
lung	L020	50	1	0.0200	0.0035	0.1050
lung	L021	5483	11	0.0020	0.0011	0.0036
lung	L022	3467	5	0.0014	0.0006	0.0034
lung	L023	3432	16	0.0047	0.0029	0.0076
lung	L024	4446	21	0.0047	0.0031	0.0072
lung	L025	6124	6	0.0010	0.0004	0.0021
lung	L026	4878	7	0.0014	0.0007	0.0030
lung	L027	7870	42	0.0053	0.0040	0.0072
lung	L028	4	0	0.0000	0.0000	0.4899
lung	L030	3027	21	0.0069	0.0045	0.0106
lung	L031	5836	14	0.0024	0.0014	0.0040
lung	L032	1	0	0.0000	0.0000	0.7935
lung	L033	3334	10	0.0030	0.0016	0.0055

source_compartment	donor.id.tag	n_cells	n_predicted	predicted_fraction	predicted_fraction_ci_low	predicted_fraction_ci_high
lung	L034	9820	63	0.0064	0.0050	0.0082
lung	L036	8773	45	0.0051	0.0038	0.0069
lung	L037	5973	17	0.0028	0.0018	0.0046
lung	L040	256	3	0.0117	0.0040	0.0339
lymph_node	L003	615	2	0.0033	0.0009	0.0118
lymph_node	L004	2435	7	0.0029	0.0014	0.0059
lymph_node	L005	2488	4	0.0016	0.0006	0.0041
lymph_node	L009	932	7	0.0075	0.0036	0.0154
lymph_node	L014	3972	7	0.0018	0.0009	0.0036
lymph_node	L024	1838	11	0.0060	0.0033	0.0107
lymph_node	L035	876	0	0.0000	0.0000	0.0044
lymph_node	L037	617	1	0.0016	0.0003	0.0091
lymph_node		889	4	0.0045	0.0018	0.0115

TCR-gene expression quadrants

source_compartment	tcr_gene_quadrant	n_cells	n_predicted	predicted_fraction	median_score
lung	TRDC+TRDV+	359	142	0.3955	0.8078
lung	TRDC+TRDV-	286	8	0.0280	0.0131
lung	TRDC-TRDV+	2665	399	0.1497	0.2769
lung	TRDC-TRDV-	139874	59	0.0004	0.0004
lymph_node	TRDC+TRDV+	19	3	0.1579	0.2497
lymph_node	TRDC+TRDV-	106	3	0.0283	0.0018
lymph_node	TRDC-TRDV+	257	35	0.1362	0.1188
lymph_node	TRDC-TRDV-	14280	2	0.0001	0.0001

TRA/TRB sequencing evidence

source_compartment	has_TRA_evidence	has_TRB_evidence	has_paired_TRA_TRB_evidence	n_cells	n_predicted	predicted_fraction
lung	False	True	False	22181	318	0.0143
lung	True	False	False	4125	53	0.0128

source_compartment	has_TRA_evidence	has_TRB_evidence	has_paired_TRA_TRB_evidence	n_cells	n_predicted	predicted_fraction
lung	True	True	True	116878	237	0.0020
lymph_node	False	True	False	2676	31	0.0116
lymph_node	True	False	False	418	4	0.0096
lymph_node	True	True	True	11568	8	0.0007

Reproducibility

Source: GEO GSE305372. Study: Fajardo-Rosas et al., Nature Immunology (2026).

Workflow: `workflows/intake/export_gse305372_cd8_model_payload.R` followed by `workflows/gdtai/run_gdtai_gse305372_cd8.py`.